ITI – RedHatAdmin 2 lab3

Mohammad Tarek Omar – Data Management

1.What is the purpose of the at command?

The at command in Linux is used to schedule tasks for one-time execution at a specific time in the future

2.How do you schedule a job to run once at 10:00 PM?

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~   mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ sudo yum install at
Last metadata expiration check: 0:45:25 ago on Sun 18 Jan 2026 02:57:14 PM EET.
Package at-3.1.23-13.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[mohammadtarek@localhost ~]$ sudo systemctl status atd
● atd.service - Deferred execution scheduler
     Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enab>
     Active: active (running) since Sun 2026-01-18 10:18:52 EET; 5h 24min ago
       Docs: man:atd(8)
   Main PID: 900 (atd)
      Tasks: 1 (limit: 10519)
     Memory: 192.0K (peak: 1.2M)
        CPU: 17ms
     CGroup: /system.slice/atd.service
             └─900 /usr/sbin/atd -f

Jan 18 10:18:52 localhost.localdomain systemd[1]: Started Deferred execution sc>
```

```
[mohammadtarek@localhost ~]$ at now+1min << END
> echo 'There is a potential to love linux' > /tmp/file
> END
warning: commands will be executed using /bin/sh
job 1 at Sun Jan 18 15:48:00 2026
[mohammadtarek@localhost ~]$ tail -f /tmp/file
tail: cannot open '/tmp/file' for reading: No such file or directory
tail: no files remaining
[mohammadtarek@localhost ~]$ sudo tail -f /tmp/file
There is a potential to love linux
```

3.How do you list pending at jobs? By using ‘atq’ command

```
[mohammadtarek@localhost ~]$ atq
5       Sun Jan 18 17:02:00 2026 a mohammadtarek
[mohammadtarek@localhost ~]$
```

4.How do you remove a scheduled at job? By using ‘atrm’ command

```
[mohammadtarek@localhost ~]$ atrm 5
[mohammadtarek@localhost ~]$
```

5.What command shows the content of an at job? ‘at -c’ command

```
DDRESS
MAIL=/var/spool/mail/mohammadtarek; export MAIL
cd /home/mohammadtarek || {
         echo 'Execution directory inaccessible' >&2
         exit 1
}
${SHELL:-/bin/sh} << 'marcinDELIMITER121a3327'
#!/bin/bash
echo 'Helloo from script running question number 9' > /tmp/file


marcinDELIMITER121a3327
[mohammadtarek@localhost ~]$
```

6.How do at.allow and at.deny work? By default there is an at.deny file only, it means that all users in this file are not allowed to use ‘at’ command, and other users can, when at.allow comes alone it means that all users in this file are allowed to use the command, and other users can not, and if there are the 2 files, at.deny and at.allow and there is a user in both files it means that at.allow has the upper hand and at.deny is useless.

7.What is the difference between at and cron? The at command in Linux is used to schedule tasks for one-time execution at a specific time in the future. Unlike cron, which is used for recurring tasks, at is ideal for ad hoc or single-use jobs.

8.How do you schedule a job to run 5 minutes from now?

```
[mohammadtarek@localhost ~]$ at now+5min << END
echo 'Okay, we love eng. Mina' >/tmp /file
END
warning: commands will be executed using /bin/sh
job 2 at Sun Jan 18 16:03:00 2026
[mohammadtarek@localhost ~]$
```

9.How do you schedule a job using a script file with at?

```
mohammadtarek@localhost:~
mohammadtarek@localhost:~    mohammadtarek@
[mohammadtarek@localhost ~]$ vim /tmp/file.sh
[mohammadtarek@localhost ~]$ chmod u+x /tmp/file.sh
[mohammadtarek@localhost ~]$
```

```
mohammadtarek@localhost:~ — vim /tmp/file.sh
mohammadtarek@localhost:~ — ...    mohammadtarek@localhost:~ — ...
#!/bin/bash
echo 'Helloo from script running question number 9' > /tmp/file
~
~
~
```

```
[mohammadtarek@localhost ~]$ vim /tmp/file.sh
You have new mail in /var/spool/mail/mohammadtarek
[mohammadtarek@localhost ~]$ at now+1min < /tmp/file.sh
warning: commands will be executed using /bin/sh
job 4 at Sun Jan 18 16:52:00 2026
[mohammadtarek@localhost ~]$ sudo tail -f /tmp/file
[sudo] password for mohammadtarek:
Helloo from script running question number 9
```

10.What daemon is responsible for executing at jobs? The at command relies on the atd daemon to execute script and enable it

11.How do you edit the current user's crontab? Crontab -e

12.How do you list scheduled cron jobs for the current user? Crontab -l

13.How do you delete all cron jobs for the current user? Crontab -r

14.Cron Syntax Questions

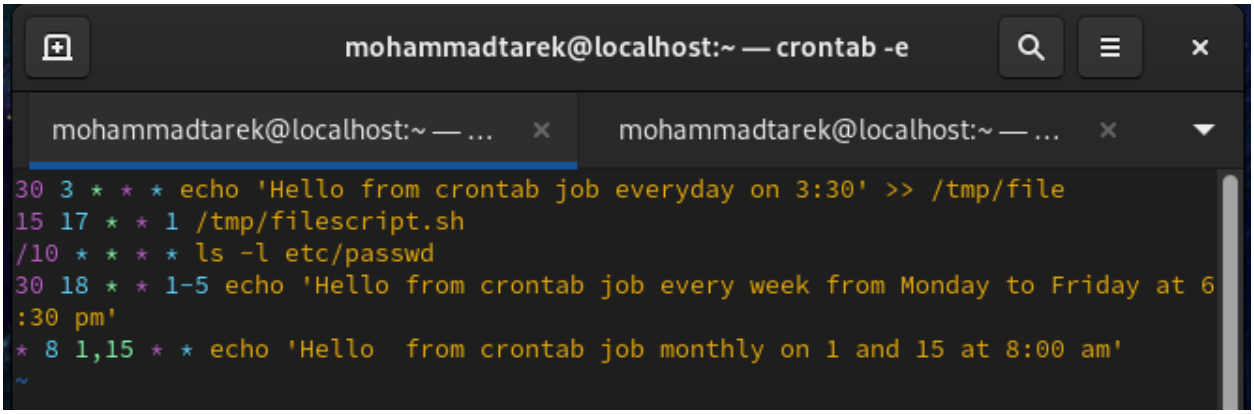
a- Schedule a job to run every day at 3:30 AM

b- Run a script every Monday at 5:15 PM

c- Execute a command every 10 minutes

d- Schedule a task to run every weekday (Mon-Fri) at 6:30 PM

e- Run a job on the 1st and 15th of each month at 8:00 AM



15.How do you restrict users from using cron? To put them into /etc/cron.deny

16.How do you check cron service status?

```
mohammadtarek@localhost:~ — sudo systemctl status crond
mohammadtarek@localhost:~ — ...   mohammadtarek@localhost:~ — ...
[mohammadtarek@localhost ~]$ sudo systemctl status cron
Unit cron.service could not be found.
[mohammadtarek@localhost ~]$ sudo systemctl status cron.service



[mohammadtarek@localhost ~]$ sudo systemctl status cron.service
Unit cron.service could not be found.
[mohammadtarek@localhost ~]$ sudo systemctl status crond
● crond.service - Command Scheduler
     Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: en>
     Active: active (running) since Sun 2026-01-18 10:18:53 EET; 6h ago
   Main PID: 902 (crond)
      Tasks: 1 (limit: 10519)
     Memory: 2.2M (peak: 4.6M)
        CPU: 785ms
     CGroup: /system.slice/crond.service
             └─902 /usr/sbin/crond -n

Jan 18 15:01:01 localhost.localdomain CROND[28942]: (root) CMD (run-parts /etc/>
Jan 18 15:01:01 localhost.localdomain run-parts[28945]: (/etc/cron.hourly) star>
Jan 18 15:01:01 localhost.localdomain run-parts[28951]: (/etc/cron.hourly) fini>
Jan 18 15:01:01 localhost.localdomain CROND[28941]: (root) CMDEND (run-parts /e>
Jan 18 16:01:01 localhost.localdomain CROND[30058]: (root) CMD (run-parts /etc/>
Jan 18 16:01:01 localhost.localdomain run-parts[30061]: (/etc/cron.hourly) star>
Jan 18 16:01:01 localhost.localdomain run-parts[30067]: (/etc/cron.hourly) fini>
```

17.How do you disable a cron job without deleting it?

```
mohammadtarek@localhost:~ — crontab -e
mohammadtarek@localhost:~ — ...   mohammadtarek@localhost:~ — ...
# 30 3 * * * echo 'Hello from crontab job everyday on 3:30' >> /tmp/file
~
~
~
```

18.How do you set environment variables in crontab? I do not know, even after searching...

19.Why might a cron job run manually but fail in crontab? Permissions, incorrect path, private path.

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Self Study

SetUID

SetGID

Sticky Bit

Self-Study Doneeee